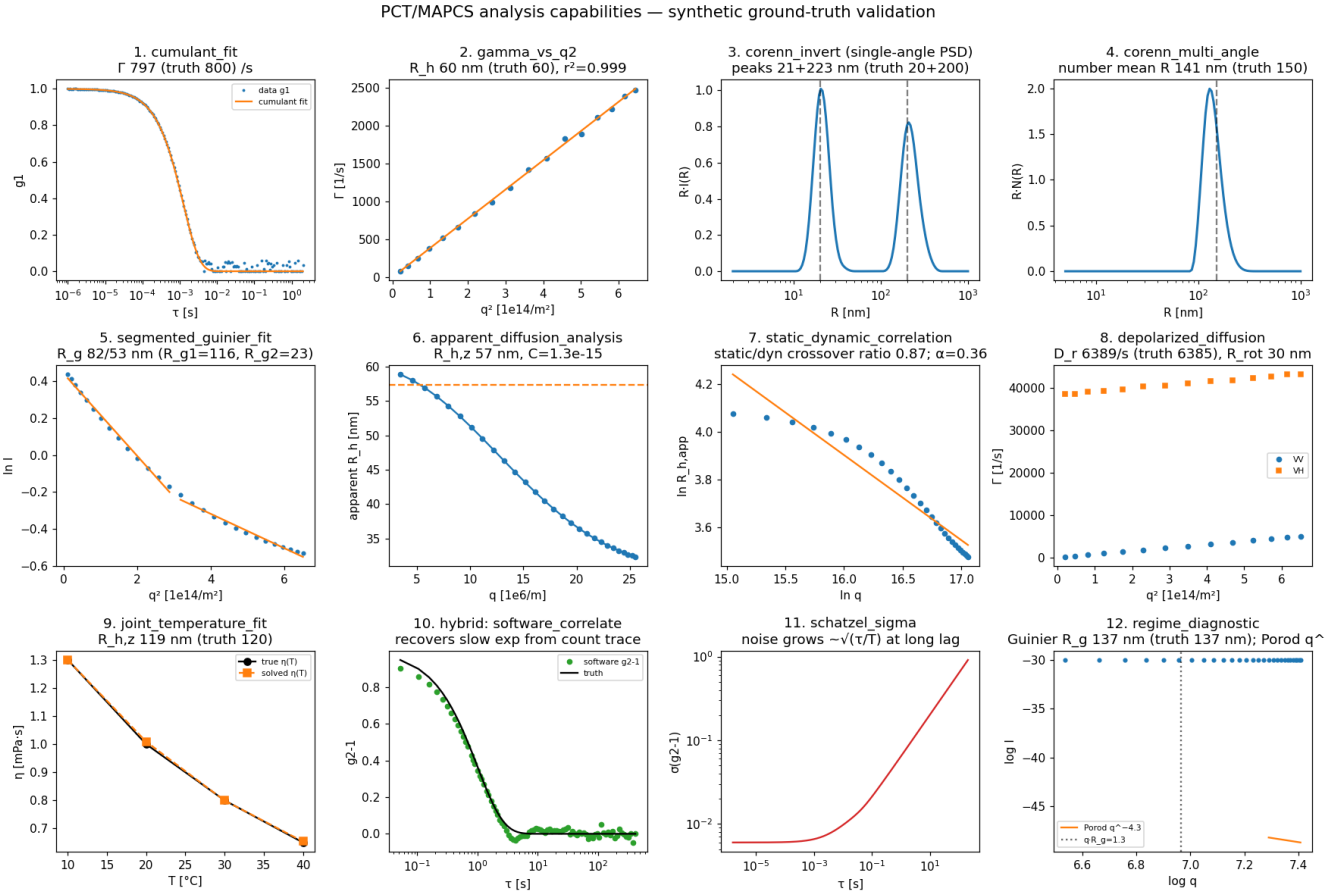
PCT / MAPCS analysis capabilities

Synthetic ground-truth validation of every analysis function in `corenn.py` and `hybrid_correlator.py`. Each was run on data generated from a known truth; the recovered values are below and visualized in `figures/capabilities.png`.



#	capability	synthetic validation
1	`cumulant_fit`	Γ=797/s (truth 800), PDI=-0.012
2	`gamma_vs_q2`	R_h=60 nm (truth 60), D=3.86e-12, r^2=0.999
3	`corenn_invert`	bimodal peaks 21+223 nm (truth 20+200), λ=2.5e-06
4	`corenn_multi_angle`	number-mean R=141 nm (truth 150), λ=4.8e-01
5	`segmented_guinier_fit`	DETECTS two length scales (ΔBIC=75 $\gg$10); effective R_g 82/53 nm (intensity-weighted low/high-q projections of populations R_g1=116, R_g2=23 nm)
6	`apparent_diffusion_analysis`	R_h,z=57 nm, C=1.3e-15, span=1.7, spearman=-1.00
7	`static_dynamic_correlation`	static-Guinier-break and dynamic-R_h-crossover q coincide (ratio 0.87); R_h,app$\propto$q^-α with α=0.36 (r^2=0.92); per-population shape ratio ρ=R_g/R_h 1.43/1.61
8	`depolarized_diffusion`	D_t$\rightarrow$R_h=30 nm (truth 30); 6D_r=38333/s (truth 38311); R_rot=30 nm
9	`joint_temperature_fit`	shared R_h,z=119 nm (truth 120); η(T) solved within 1% of truth
10	`hybrid_correlator`	software correlation of count trace recovers g2-1 (RMS 0.029 vs truth)
11	`schatzel_sigma`	Schätzel weighting: σ flat at short lag, grows ~$\sqrt{(\tau/T)}$ at long lag
12	`regime_diagnostic`	PSD (R=182 nm) $\rightarrow$ q·R_g spans 0.5–3.6: spans Guinier (low-q) and Porod (high-q); low-q Guinier R_g=137 nm (truth 137), high-q I$\propto$q^-4.3 (Porod q^-4 envelope)

Notes

- The **forward model is primary**: `corenn_invert` / `corenn_multi_angle` fit the full `g2(q,`τ`)` for the size distribution; the cumulant Γ`(q)`, apparent `R_h(q)`, and the static `I(q)` regime (Guinier vs Porod, anticipated from the PSD via `regime_diagnostic`) are **diagnostic projections** of that model, not the model itself.
- Γ`(q)` alone is information-poor: it supports a two-population *summary* but cannot resolve >2 populations or a continuous distribution — that requires the full decay (`corenn_invert` / `corenn_multi_angle`).
- The hybrid hardware+software correlator is opt-in (`load_measurement_replicates(dir, hybrid=True)`); it extends the lag range for slow-mode analysis but is not used for the cross-angle inversion.